Stoichiometry

30 questions: 15 multiple choice, 15 structured.

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1 Core Formulae [1]

Limestone is mainly calcium carbonate. Which formula represents calcium carbonate?

(A) Ca_2CO_3

(B) $CaCO_2$

(C) $CaCO_3$

(D) $Ca(CO_3)_2$

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2 Core Formulae [1]

What does the molecular formula of a compound show?

(A) The number and type of each atom in one molecule of the compound

(B) The simplest whole-number ratio of atoms in the compound

(C) The charges on the ions in one formula unit of the compound

(D) The percentage by mass of each element in the compound

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3 Core Formulae [1]

Magnesium ribbon burns in air to form magnesium oxide. Which symbol equation, including state symbols, correctly represents this reaction?

(A) $Mg(s) + O_2(g) \rightarrow MgO_2(s)$

(B) $Mg(s) + O(g) \rightarrow MgO(s)$

(C) $2Mg(s) + O_2(g) \rightarrow MgO(s)$

(D) $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

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4 Extended Formulae [1]

A compound has the molecular formula C_4H_{10}. What is its empirical formula?

A C_4H_{10}

B C_2H_5

C CH_2

D C_2H_{10}

5 Extended Formulae [1]

Aluminium ions, Al^{3+}, and sulfate ions, SO_4^{2-}, combine to form an ionic compound. What is the formula of this compound?

A $Al_2(SO_4)_3$

B $AlSO_4$

C Al_2SO_4

D $Al_3(SO_4)_2$

6 Extended Formulae [1]

Aqueous silver nitrate is added to aqueous sodium chloride, forming a white precipitate. Which ionic equation, with state symbols, correctly represents this reaction?

A $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

B $Ag^+(aq) + NO_3^-(aq) \rightarrow AgNO_3(s)$

C $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

D $2Ag^+(aq) + Cl^-(aq) \rightarrow Ag_2Cl(s)$

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7 Core Formulae [4]

Sodium reacts with water to produce sodium hydroxide and hydrogen gas.

(a) Give the formula of each of the following:

(i) sodium hydroxide [1]

(ii) hydrogen gas [1]

(b) Write a balanced symbol equation for this reaction, including state symbols. [2]

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8 Core Formulae [4]

The table shows the number of each type of atom present in one molecule of three compounds, P, Q and R.

Compound	Carbon atoms	Hydrogen atoms	Oxygen atoms
P	1	4	0
Q	2	6	1
R	1	2	1

(a) Deduce the molecular formula of compound P. [1]

(b) Deduce the molecular formula of compound Q. [1]

(c) Deduce the molecular formula of compound R. [1]

(d) State the meaning of the term *molecular formula*. [1]

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9 **Core** Formulae [4]

Calcium carbonate is a solid. When it is heated strongly in an open container, it decomposes into calcium oxide, which is also a solid, and carbon dioxide gas.

(a) Write a word equation for this reaction. [1]

(b) Write a balanced symbol equation for this reaction, including state symbols. [2]

(c) Explain, using the equation, why the total mass of solid remaining decreases when calcium carbonate is heated in an open container. [1]

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10 **Extended** Formulae [4]

A hydrocarbon molecule contains 6 carbon atoms and 12 hydrogen atoms.

(a) State the meaning of the term *empirical formula*. [1]

(b) Give the molecular formula of this hydrocarbon. [1]

(c) Deduce the empirical formula of this hydrocarbon. [1]

(d) A different hydrocarbon has the empirical formula CH_2 and its molecule contains 4 carbon atoms. Deduce its molecular formula. [1]

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11 **Extended** Formulae [5]

The table gives the formula and charge of four ions.

Ion	Formula	Charge
Barium	Ba	2+
Sulfate	SO_4	2−
Chloride	Cl	1−
Sodium	Na	1+

(a) Deduce the formula of barium sulfate. [1]

(b) Deduce the formula of sodium chloride. [1]

(c) Aqueous barium chloride is mixed with aqueous sodium sulfate. A white precipitate of barium sulfate forms, leaving sodium chloride in solution. Write a balanced symbol equation for this reaction, including state symbols. [3]

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12 **Extended** Formulae [5]

Dilute hydrochloric acid reacts with aqueous sodium hydroxide in a neutralisation reaction, producing water and aqueous sodium chloride.

(a) Write a full balanced symbol equation for this reaction, including state symbols. [2]

(b) Identify the two spectator ions in this reaction. [1]

(c) Write the ionic equation for this neutralisation reaction, including state symbols. [2]

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13 **Core** Relative masses of atoms and molecules [1]

Which statement correctly defines relative atomic mass, Ar?

A The mass of one atom of an element measured in grams

B The average mass of the isotopes of an element compared with 1/12 of the mass of an atom of carbon-12

C The number of protons and neutrons in the nucleus of an atom

D The mass of one mole of atoms of an element

14 Core Relative masses of atoms and molecules [1]

Using Ar: C = 12, O = 16, Ca = 40, what is the relative formula mass, Mr, of calcium carbonate, $CaCO_3$?

A 68

B 84

C 116

D 100

15 Core Relative masses of atoms and molecules [5]

Ar: H = 1, C = 12, N = 14, O = 16.

(a) Calculate the relative molecular mass, Mr, of carbon dioxide, CO_2. [1]

(b) Calculate the relative molecular mass, Mr, of glucose, $C_6H_{12}O_6$. [2]

(c) Calculate the relative formula mass, Mr, of ammonium nitrate, NH_4NO_3. [2]

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16 Core Relative masses of atoms and molecules [4]

In the reaction $2Mg + O_2 \rightarrow 2MgO$, 24 g of magnesium reacts completely with oxygen to form 40 g of magnesium oxide (relative formula mass of MgO = 40).

(a) Calculate the mass of oxygen that reacts with 24 g of magnesium. [1]

(b) Calculate the mass of magnesium oxide formed when 9.6 g of magnesium is completely oxidised, using simple proportion. [2]

(c) Calculate the mass of oxygen needed to react completely with 9.6 g of magnesium. [1]

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17 Core Relative masses of atoms and molecules [4]

Copper carbonate decomposes on heating: $CuCO_3 \rightarrow CuO + CO_2$

The relative formula masses are: $CuCO_3 = 124$, $CuO = 80$, $CO_2 = 44$.

(a) Show that these three relative formula masses are consistent with conservation of mass in this equation. [1]

(b) 3.1 g of copper carbonate is heated until it decomposes completely. Calculate the mass of copper oxide formed. [2]

(c) Calculate the mass of carbon dioxide gas released in part (b). [1]

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18 Core The mole and the Avogadro constant [1]

Which pair of units can both be used to express the concentration of a solution?

A g/dm^3 and mol/dm^3

B g/cm^3 and mol/g

C mol/dm^3 and dm^3/mol

D g/mol and mol/dm^3

19 Extended The mole and the Avogadro constant [1]

What is the value of the Avogadro constant, and what does it represent?

A 6.02 × 10^23; the number of moles in one gram of a substance

B 6.02 × 10^-23; the number of particles in one mole of a substance

C 6.02 × 10^23; the number of particles in one mole of a substance

D 6.02 × 10^23; the mass, in grams, of one mole of a substance

20 Extended The mole and the Avogadro constant [1]

Ar: Na = 23. What amount, in moles, is present in 11.5 g of sodium?

A 0.5 mol

B 0.05 mol

C 2.0 mol

D 5.0 mol

21 Extended The mole and the Avogadro constant [1]

The molar gas volume is 24 dm^3 at room temperature and pressure (r.t.p.). What is the volume occupied by 0.25 mol of a gas at r.t.p.?

A 24 dm^3

B 96 dm^3

C 0.25 dm^3

D 6 dm^3

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22 **Extended** The mole and the Avogadro constant [1]

0.25 mol of solute is dissolved in 250 cm^3 of solution. What is the concentration of this solution, in mol/dm^3?

A 0.1 mol/dm^3

B 1.0 mol/dm^3

C 4.0 mol/dm^3

D 1000 mol/dm^3

23 **Extended** The mole and the Avogadro constant [1]

A compound contains 82.4% nitrogen and 17.6% hydrogen by mass. Ar: N = 14, H = 1. What is the empirical formula of this compound?

A NH

B NH_2

C NH_3

D N_3H

24 **Extended** The mole and the Avogadro constant [1]

A student expected to obtain 12.0 g of a salt from a reaction but actually obtained 9.6 g. What is the percentage yield?

A 20%

B 80%

C 96%

D 125%

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25 **Extended** The mole and the Avogadro constant [5]

Ar: Na = 23, O = 16, H = 1.

A solution of sodium hydroxide has a concentration of 8.0 g/dm^3.

(a) Calculate the relative formula mass, Mr, of sodium hydroxide, NaOH. [1]

(b) Calculate the concentration of this solution in mol/dm^3. [2]

(c) Calculate the amount, in moles, of sodium hydroxide present in 250 cm^3 of this solution. [2]

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26 **Extended** The mole and the Avogadro constant [5]

Ar: Mg = 24.

(a) Calculate the amount, in moles, of magnesium atoms in 3.0 g of magnesium. [2]

(b) Calculate the number of magnesium atoms in this 3.0 g sample. [2]

(c) Calculate the mass of 0.60 mol of magnesium atoms. [1]

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27 **Extended** The mole and the Avogadro constant [7]

Ar: Mg = 24, H = 1, Cl = 35.5.

Magnesium reacts with hydrochloric acid: $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

2.4 g of magnesium is added to 120 cm^3 of hydrochloric acid of concentration 2.0 mol/dm^3.

(a) Calculate the amount, in moles, of magnesium used. [1]

(b) Calculate the amount, in moles, of hydrochloric acid used. [2]

(c) Determine which reactant is in excess. Show your working. [2]

(d) Calculate the maximum mass of magnesium chloride, $MgCl_2$, that could form. [2]

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28 **Extended** The mole and the Avogadro constant [4]

Ar: Na = 23, C = 12, O = 16.

Excess dilute sulfuric acid reacts with 5.3 g of sodium carbonate:

$Na_2CO_3(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + H_2O(l) + CO_2(g)$

(a) Calculate the amount, in moles, of sodium carbonate reacted. [1]

(b) Calculate the amount, in moles, of carbon dioxide gas produced. [1]

(c) Calculate the volume of carbon dioxide gas produced, measured at r.t.p. [2]

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29 **Extended** The mole and the Avogadro constant [6]

In a titration, 25.0 cm^3 of a sodium hydroxide solution of unknown concentration was exactly neutralised by 22.5 cm^3 of hydrochloric acid of concentration 0.100 mol/dm^3.

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

(a) Calculate the amount, in moles, of hydrochloric acid used. [1]

(b) Calculate the amount, in moles, of sodium hydroxide that reacted. [1]

(c) Calculate the concentration of the sodium hydroxide solution, in mol/dm^3. [2]

(d) Ar: Na = 23, O = 16, H = 1. Calculate the concentration of the sodium hydroxide solution in g/dm^3. [2]

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30 **Extended** The mole and the Avogadro constant [5]

A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass. Its relative molecular mass, Mr, is 42.

Ar: C = 12, H = 1.

(a) Calculate the empirical formula of this hydrocarbon. Show your working. [3]

(b) Calculate the molecular formula of this hydrocarbon. [2]

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